BI 1.0 & Data Warehousing

The turbulent evolution of data analysis

Matteo Francia <m.francia@unibo.it>

BBS
MULTIPLYING OPPORTUNITIES

BOLOGNA BUSINESS SCHOOL
UNIVERSITÀ DI BOLOGNA
A.D. 1088

Glossary

Database

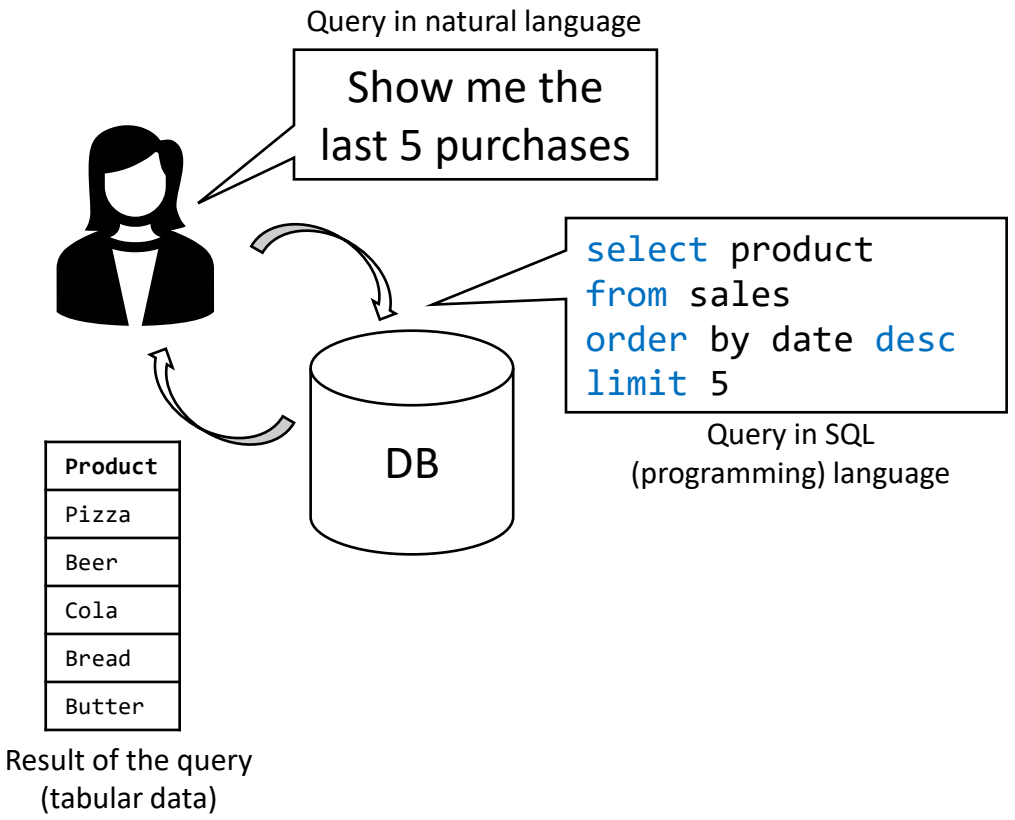
- Any collection of data that is specially organized for rapid search and retrieval by a computer
- Relational databases organize data as tables

Operational database

- A system that efficiently manages and processes real-time data for day-to-day business operations

Query

- A *database query* can be defined as a request for data from a database
- Usually, it is expressed in a structured query language such as SQL



Query workloads

Imagine a supermarket.

OLTP (Online Transaction Processing) is like the cash register where you buy things.

- When you pick up some milk and go to the checkout, the system records your purchase.
- It's all about day-to-day operations — sales, refunds, payments.
- OLTP is fast because it needs to handle many people buying things at the same time.

OLAP (Online Analytical Processing) is like the manager's office, where they analyze all purchases.

- The manager looks at data from many days to answer questions like:
    - What products are selling best?
    - What times are busiest?
    - Should we order more apples next week?
- OLAP is slower, but it's okay because it's about studying and making decisions, not quick sales.

Query workloads

Key idea:

- OLTP = Quickly saving and updating information when people do everyday activities.
- OLAP = Looking at a lot of information to find patterns and make smart decisions.

In short:

- OLTP = Doing everyday work (fast transactions)
- OLAP = Studying the big picture (deep analysis)

Examples of OLTP queries

- A retailer uses a point-of-sale system to complete transactions online and in-store.
  OLTP system processes each transaction and creates a database of the information for each transaction.
  It then sends a request to the customer's credit card company, which approves the charge to the card.
  The system records the payment and deducts one item from the store's inventory.
  This transaction follows all the rules of a successful OLTP system and helps the business process payments quicker and manage its inventory more efficiently.
- A couple has a joint account at their bank, and each person attempts to withdraw the full balance of $4,502.34 on the same day at the same time.
  The OLTP system fails the transactions for violating the concurrence rule.
  The couple has to withdraw money from the same bank account at separate times for the transaction to process successfully.
  Otherwise, the bank might provide the full balance for each party, creating a deficit of $4,502.34 in the account.

Examples of OLAP queries

- “What is the relationship between performance of shares of PC manufacturers and quarterly profits along the last 5 years?”
- “What are the types of orders that will maximize revenues?”

ICT in companies

Up to some years ago, the main goal of databases in companies has been that of storing operational data, i.e., data generated by operations carried out within business processes

Computer science was seen as a subsidiary discipline that makes information management faster and cheaper, but does not create profits in itself

The evolution of information systems

- The role of computer science in companies has radically changed since the early 70's. ICT systems turned from simple tools to improve process efficiency into key factors of company organizations capable of deeply impacting on the structure of business processes

The twofold role of computer science

Auxiliary technology to manage the company information system

Organizational discipline that impacts on business processes, services, and company structure

The new role of computer science in decision making

An exponential increase in operational data has made computers the only tools suitable for providing data for decision-making performed by business managers

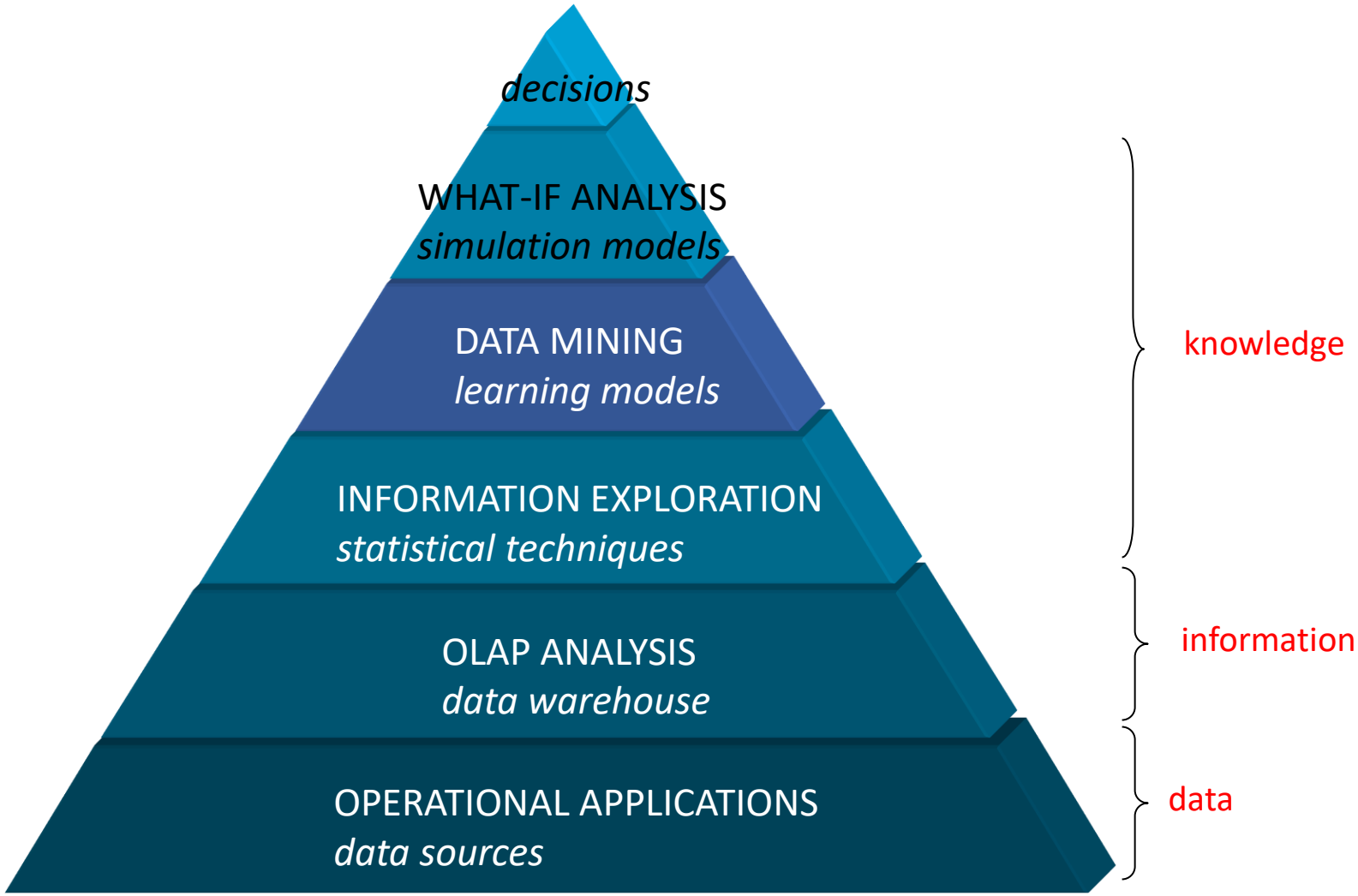
The massive use of techniques for analyzing enterprise data made information systems a key factor to achieve business goals

Business intelligence

A set of tools and techniques that enable a company to transform its business data into timely and accurate information for the decisional process

- Business intelligence systems are used by decision makers to get a comprehensive knowledge of the business and of the factors that affect it, as well as to define and support their business strategies
- The goal is to enable data-based decisions aimed at gaining competitive advantage, improving operative performance, responding more quickly to changes, increasing profitability and, in general, creating added value for the company

The BI1.0 pyramid



Data Warehousing

From data to information

Information assets are immensely valuable to any enterprise, and because of this, these assets must be properly stored and readily accessible when they are needed

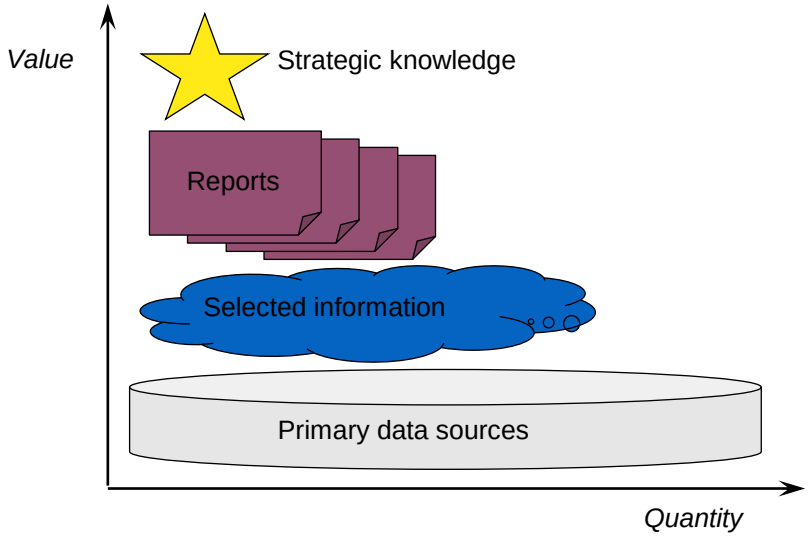
However, the availability of too much data makes the extraction of the most important information difficult, if not impossible

~~data = information~~

From data to information

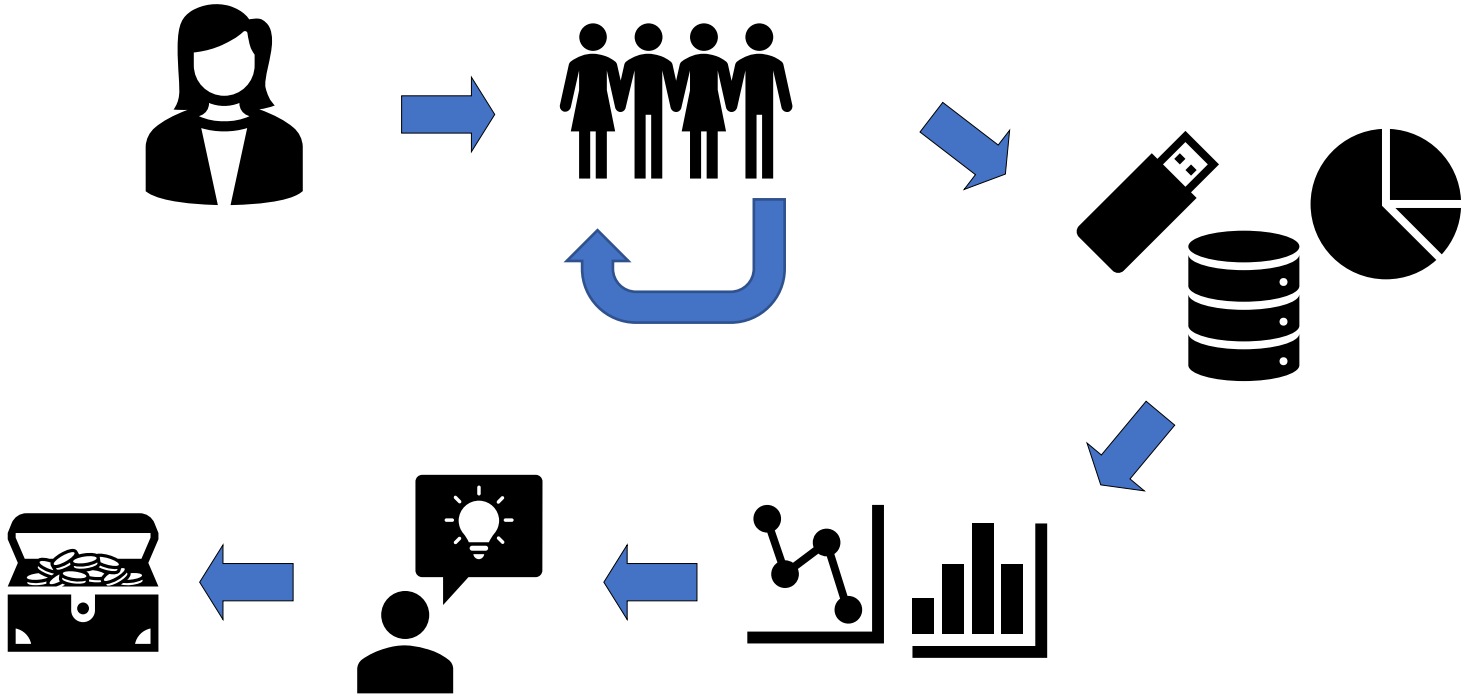
A company must be have quick access to all the information needed for decision making

Strategic knowledge is extracted from the huge amount of operational data stored in enterprise databases, through a selection and aggregation process



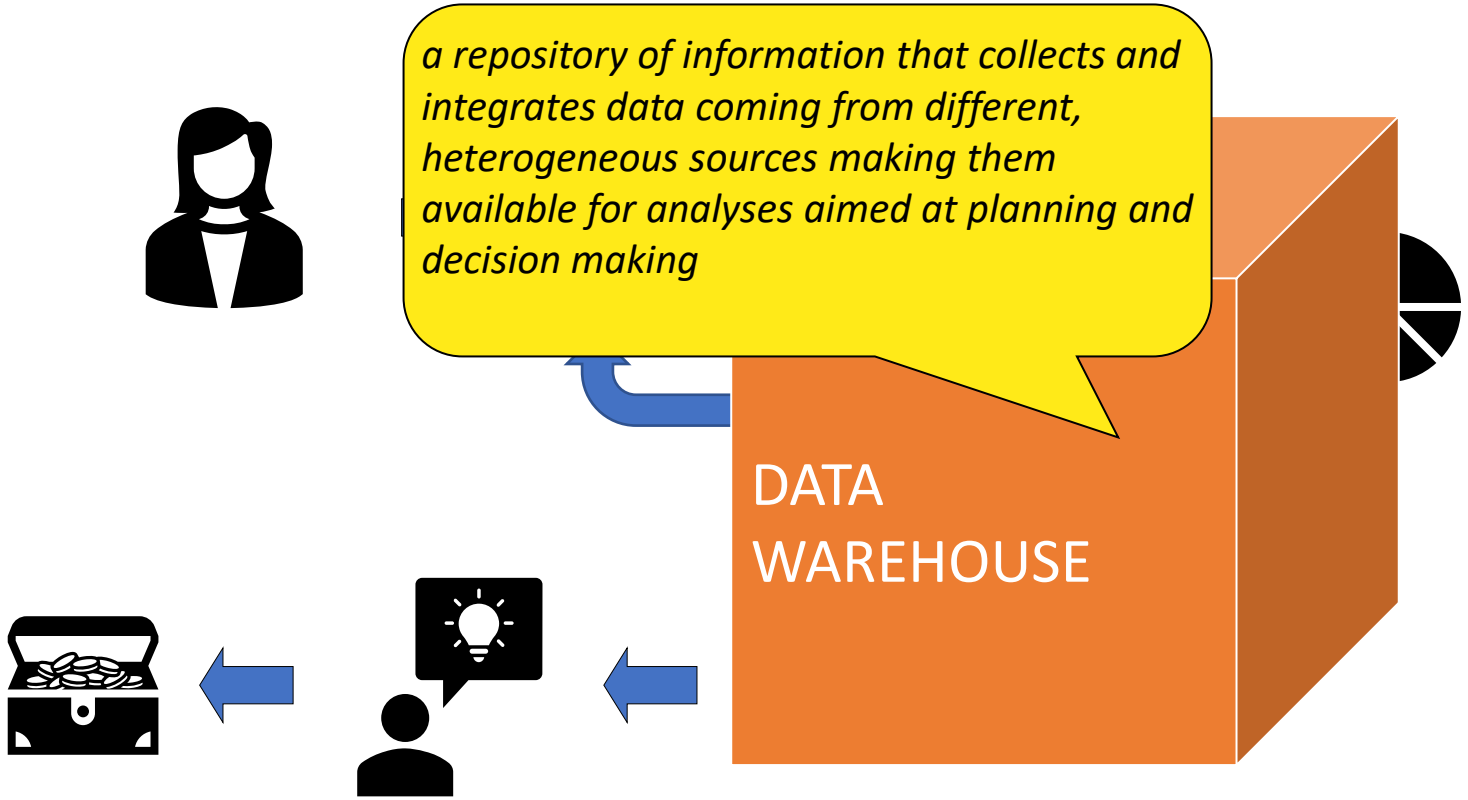
A typical scenario...

... is that of a large company, with several branches, whose managers wish to quantify and evaluate the contribution given from each branch to the global profit



A typical scenario...

... is that of a large company, with several branches, whose managers wish to quantify and evaluate the contribution given from each branch to the global profit



Queries

OLTP:

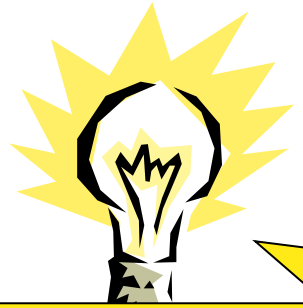
- They execute transactions that generally read/write a small number of tuples from/to many tables connected by simple relations
- The essential workload core is “frozen” in application programs, and ad hoc data queries are occasionally run for data maintenance

OLAP:

- Dynamic, multidimensional analyses that need to scan a huge amount of records to process a set of numeric data summing up the performance of an enterprise
- Interactivity is an essential property for analysis sessions, so the actual workload constantly changes as time goes by

OLTP and OLAP

A mix of analytical queries with transactional routine queries inevitably slows down the system, and this does not meet the needs of users of both types of queries



separate *online analytical processing (OLAP)* from *online transactional processing (OLTP)* by creating a new repository that integrates data from various sources and then makes data available for analysis and evaluation aimed at decision-making processes

Data Warehousing:

A collection of methods, techniques, and tools used to support *knowledge workers* (senior managers, directors, managers, and analysts) to conduct data analyses that help with performing decision-making processes and improving information resources

User claims

- *We have heaps of data, but we cannot access them!*
- *How can people playing the same role achieve substantially different results?*
- *We want to select, group, and manipulate data in every possible way!*
- *Show me just what matters!*
- *Everyone knows that some data is wrong!*

R. Kimball, The Data Warehouse Toolkit

Requirements for the warehousing process

Accessibility to users not very familiar with IT and data structures

Integration of data on the basis of a standard enterprise model

Query flexibility to maximize the advantages obtained from the existing information

Information conciseness allowing for target-oriented and effective analyses

Multidimensional representation giving users an intuitive and manageable view of information

Correctness and completeness of integrated data

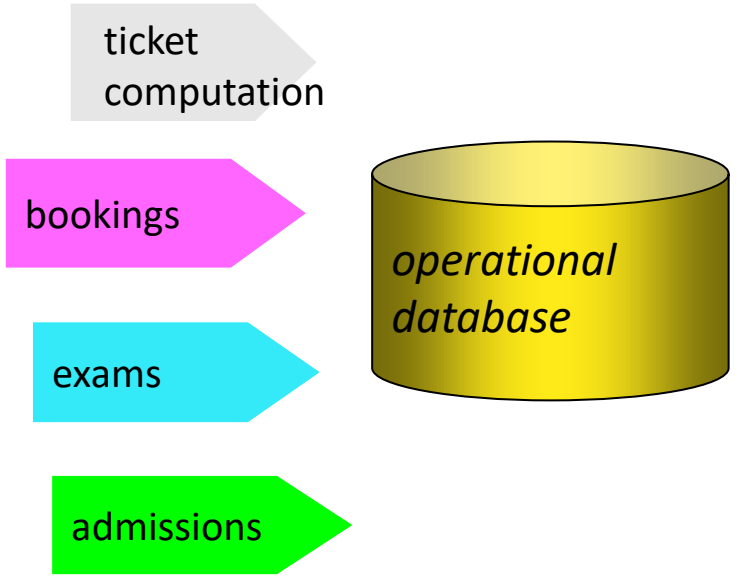
The Data Warehouse

In the middle of this process, a data warehouse is a data repository that fulfills the requirements

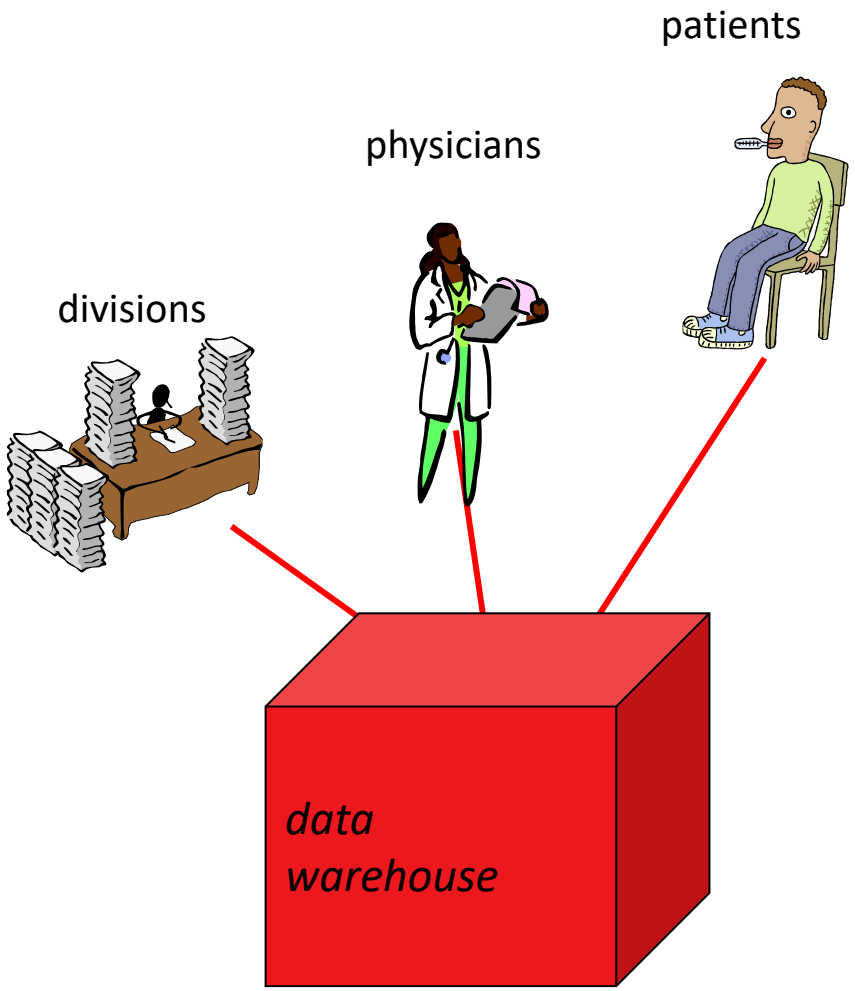
➢*A data warehouse is a collection of data that supports decision-making processes. It provides the following features:*

- *It is subject-oriented;*
- *It is integrated and consistent;*
- *It shows its evolution over time and it is not volatile*

...subject-oriented



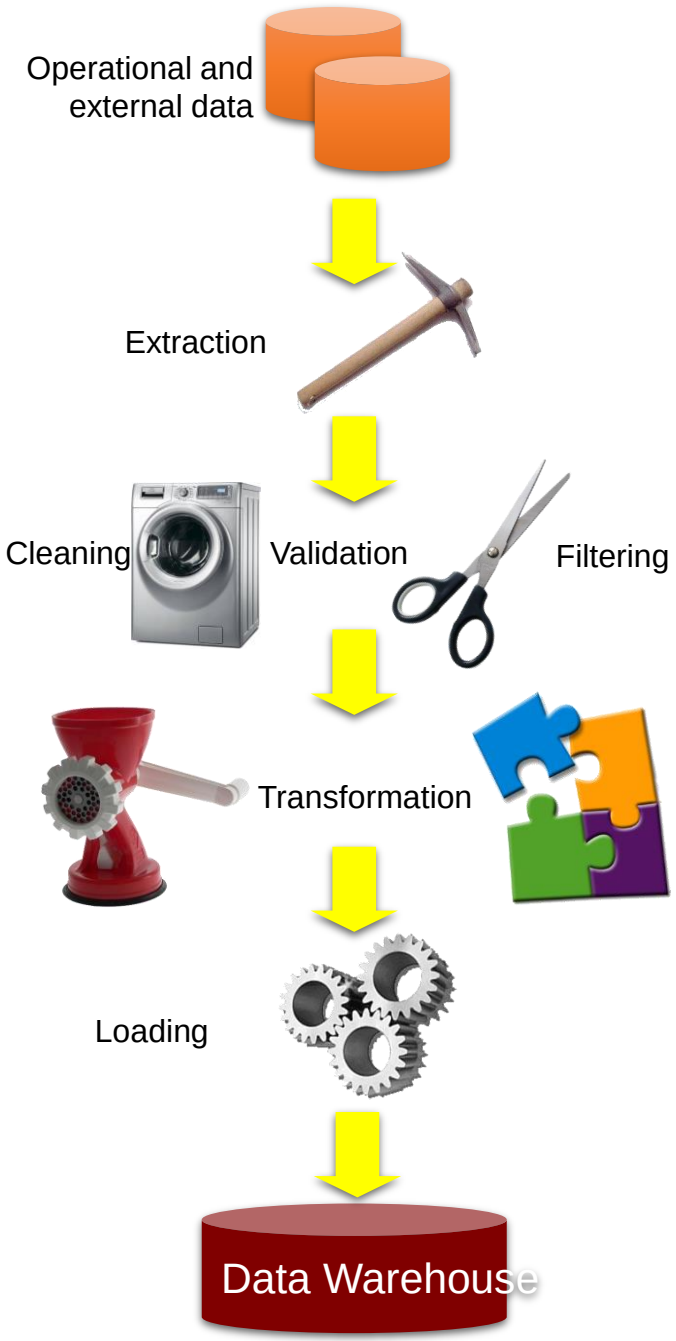
application-oriented



subject-oriented

...integrated and consistent

Data warehouses take advantage of multiple data sources, such as data extracted from production and then stored to enterprise databases, or even data from a third party's information systems. A data warehouse should provide a unified view of all the data.



...shows its temporal evolution

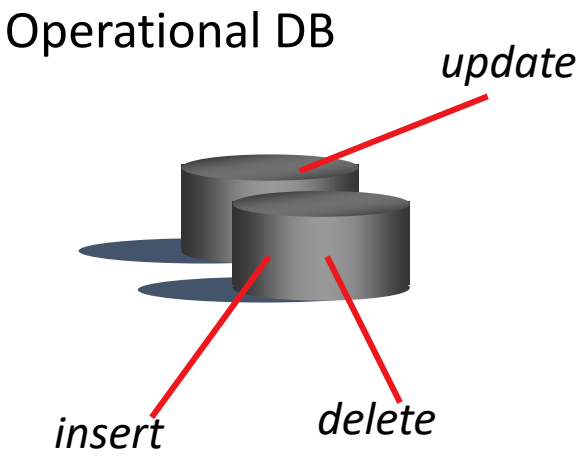
Operational DB

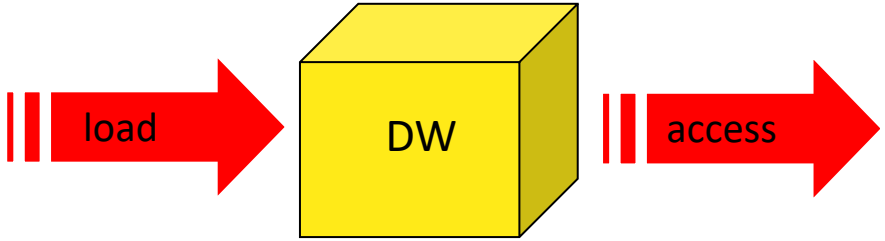
Limited historical content, time is often not part of the keys, data are updated

DW

Rich historical content, time is part of the keys, a snapshot of data taken at a given time cannot be updated

...non volatile





Huge data volumes:
from 50 GBs to some TBs
in a few years

- no need for advanced transaction management techniques required by operational applications
- key problems are query-throughput and resilience

Summarizing

	Operational DBs	***Data warehouses***
users	thousands	hundreds
workload	predefined transactions	*ad hoc* analysis queries
access	to hundreds of records, read and write	to millions of records, mostly read-only
goal	application-dependent	decision support
data	elementary, numeric and alphanumeric	aggregated, mostly numeric
lata integration	application-based	subject-based
quality	in terms of integrity	in terms of consistency
temporal span	current data	current and historical data
update	continuous	periodic
model	normalized	multidimensional
optimization	for OLTP accesses on a fraction of database	for OLAP accesses on a large part of database

Towards the multidimensional model

- “What business were registered last year for each region and each product category?”
- “What is the relationship between performance of shares of PC manufacturers and quarterly profits along the last 5 years?”
- “What are the types of orders that will maximize revenues?”
- “Which of two new therapies is more effective for decreasing the average length of hospitalizations?”
- “What is the relationship between the profits made with shipments of less than 10 items and those made with shipments of more than 10 items?”

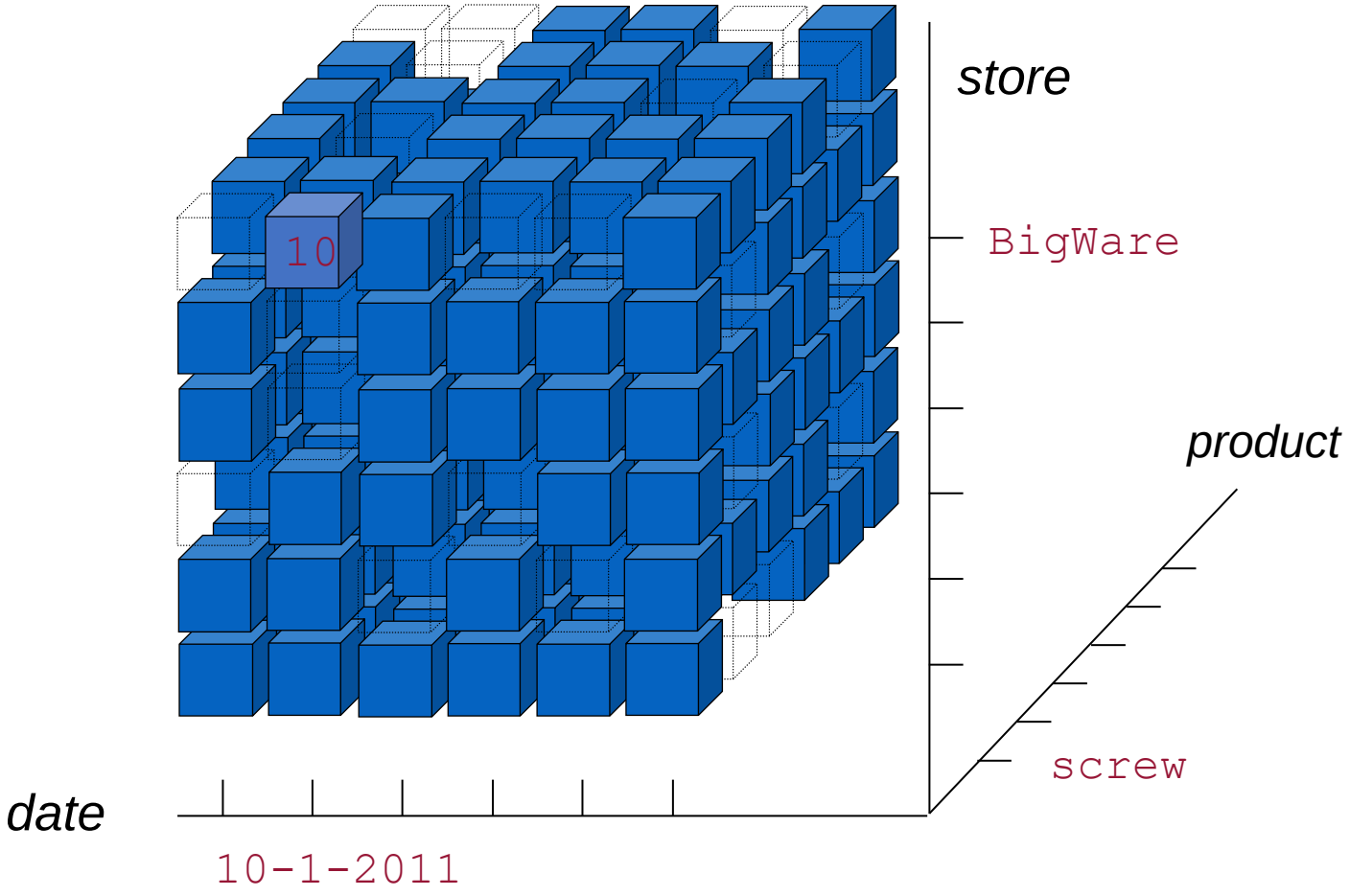
The multidimensional model

It is the key for representing and querying information in a DW

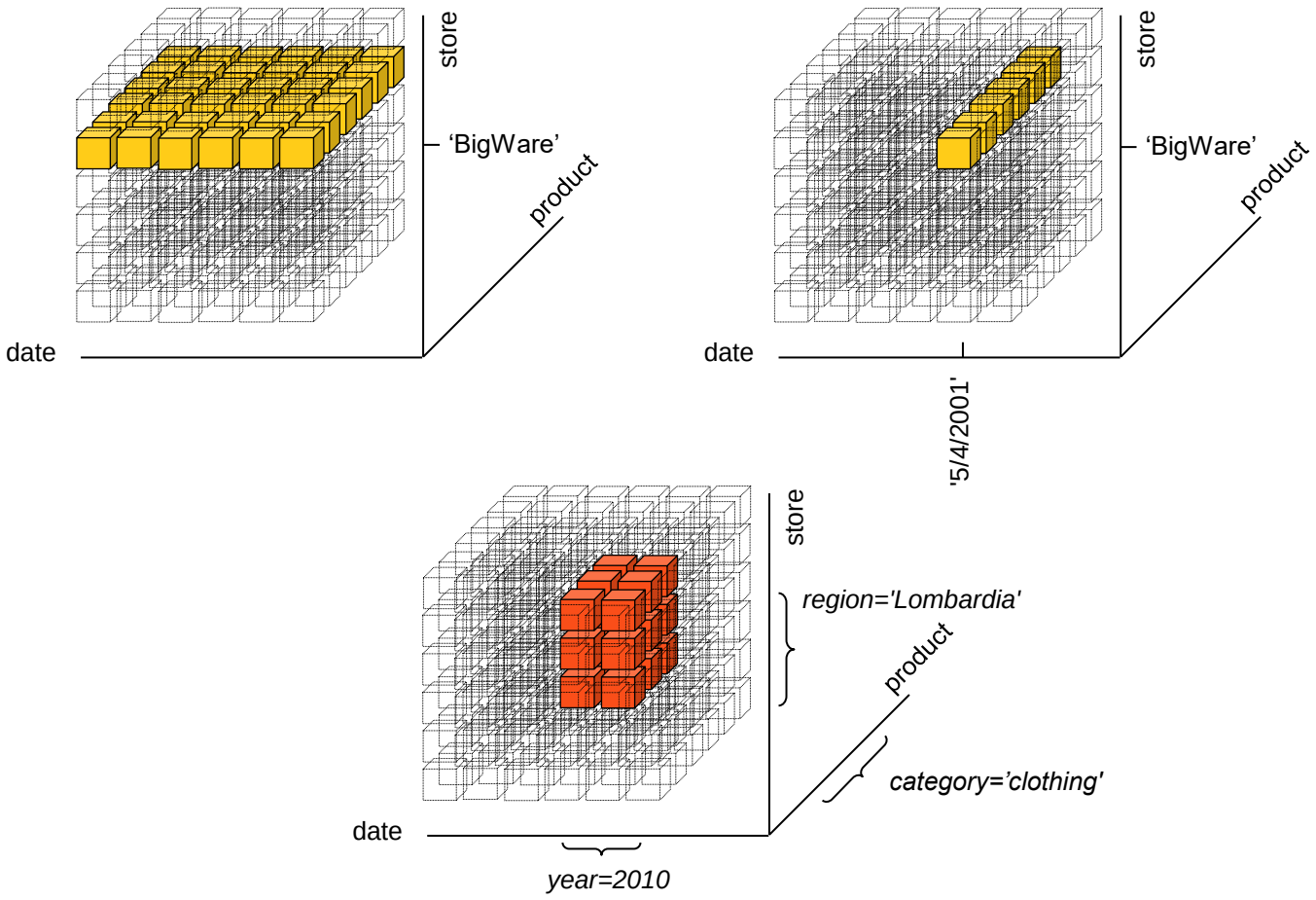
Facts of interest are represented in *cubes* where:

- each cell stores numerical *measures* that quantify the fact from different points of view;
- each axis is a *dimension* for analyzing measure values;
- each dimension can be the root of a *hierarchy* of attributes used to aggregated measure values

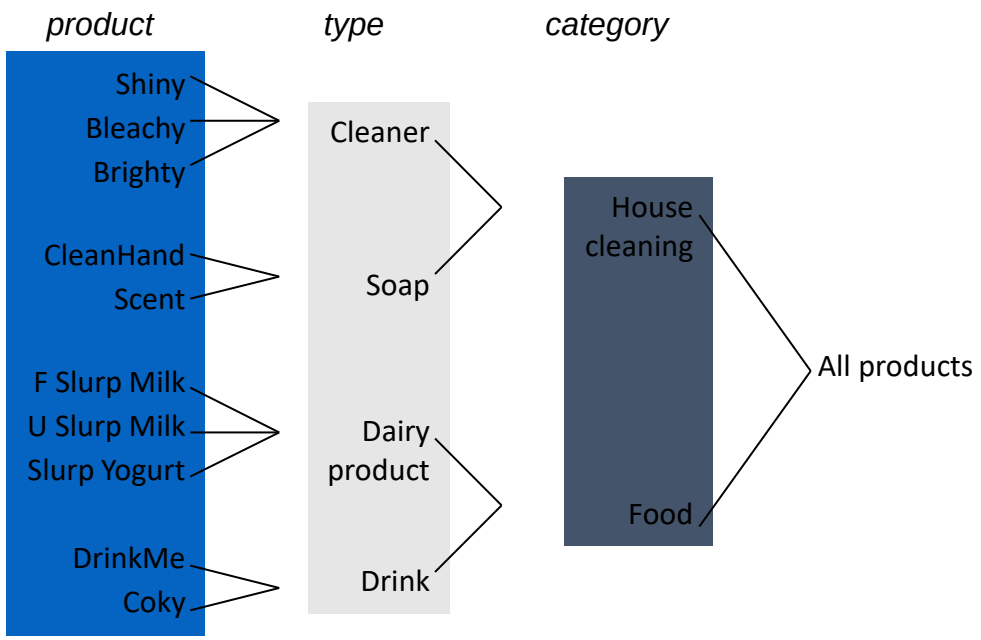
The Sales cube

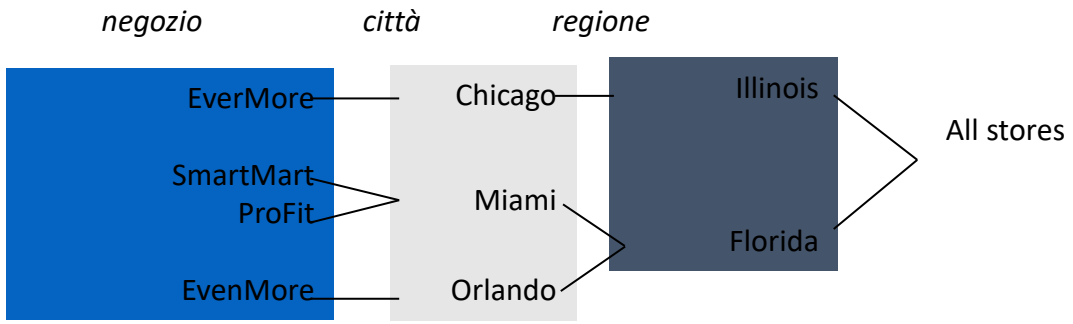


Slicing and dicing

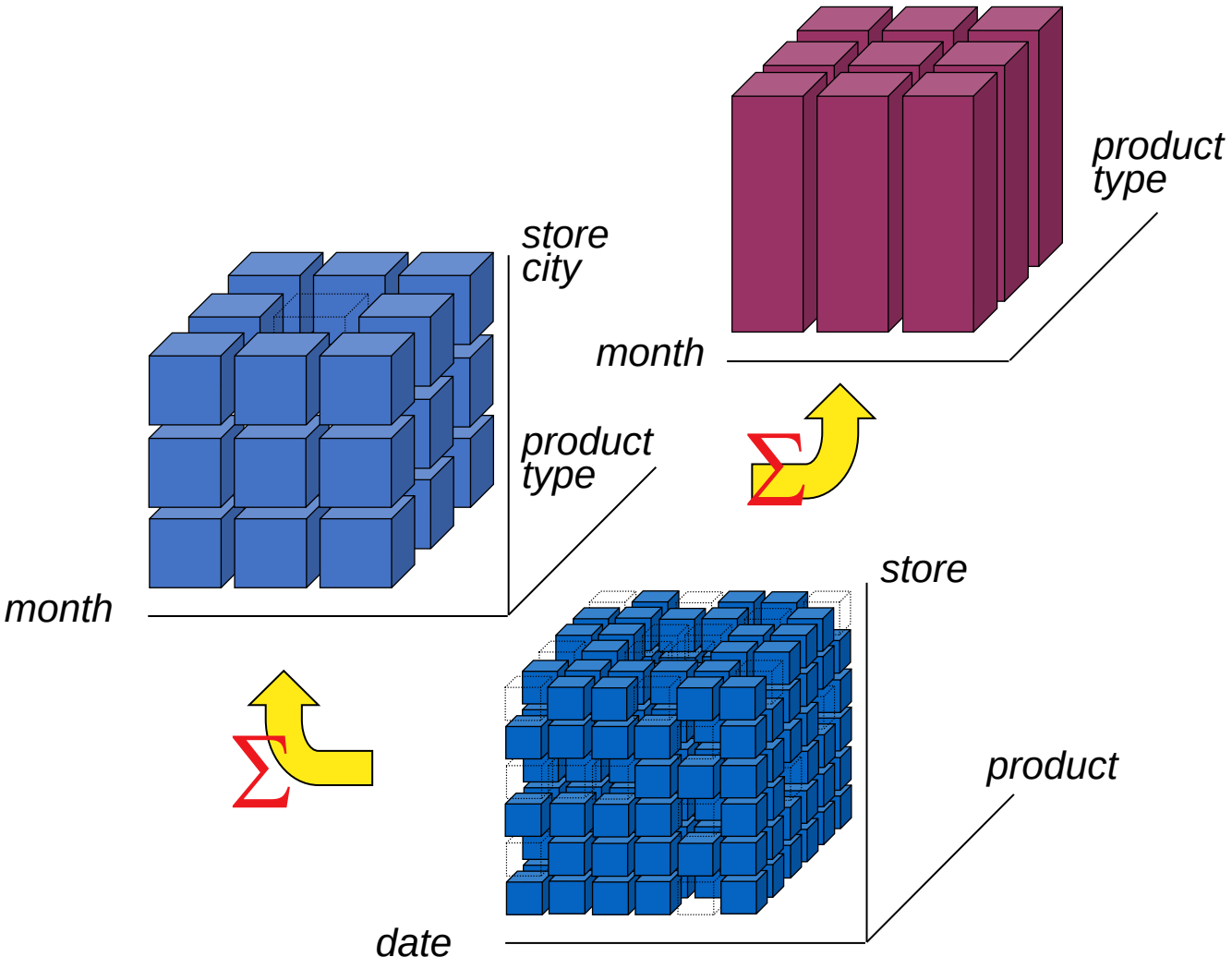


Hierarchies





Aggregation



Aggregation

	BigWare1	BigWare2	NotOnlyJelly
1/1/2000	-	-	-
2/1/2000	10	15	5
3/1/2000	20	–	5
.....	.....	.....	.....
1/1/2001	-	-	-
2/1/2001	15	10	20
3/1/2001	20	20	25
.....	.....	.....	.....
1/1/2002	-	-	-
2/1/2002	20	8	25
3/1/2002	20	12	20
.....	.....	.....	.....

	BigWare1	BigWare2	NotOnlyJelly
January 2000	200	180	150
February 2000	180	150	120
March 2000	220	180	160
.....	.....	.....	.....
January 2001	350	220	200
February 2001	300	200	250
March 2001	310	180	300
.....	.....	.....	.....
January 2002	380	200	220
February 2002	310	200	250
March 2002	300	160	280
.....	.....	.....	.....

	BigWare1	BigWare2	NotOnlyJelly
2000	2400	2000	1600
2001	3200	2300	3000
2002	3400	2200	3200

	BigWare1	BigWare22	NotOnlyJelly
Totale:	9000	6500	7800

Data analysis techniques

Once data have been cleaned, integrated, and transformed, users must be enabled to take maximum advantage from the resulting information assets

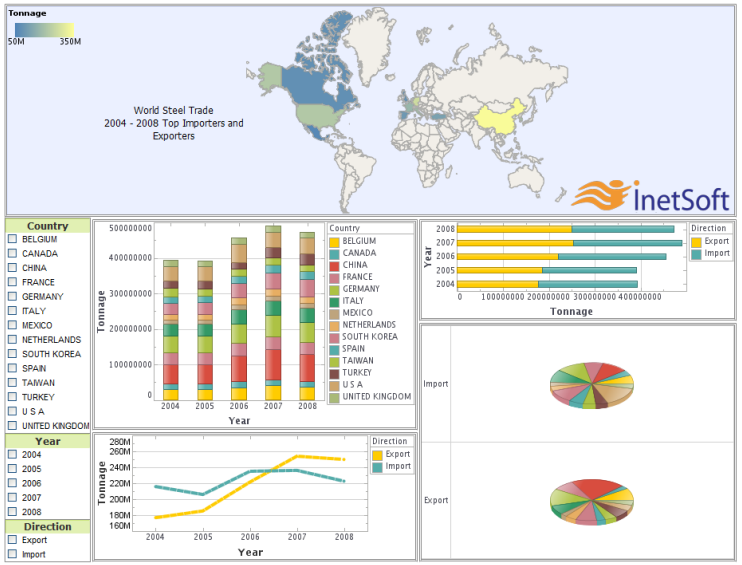
There are two different approaches for querying data warehouses, supported by different types of tools:

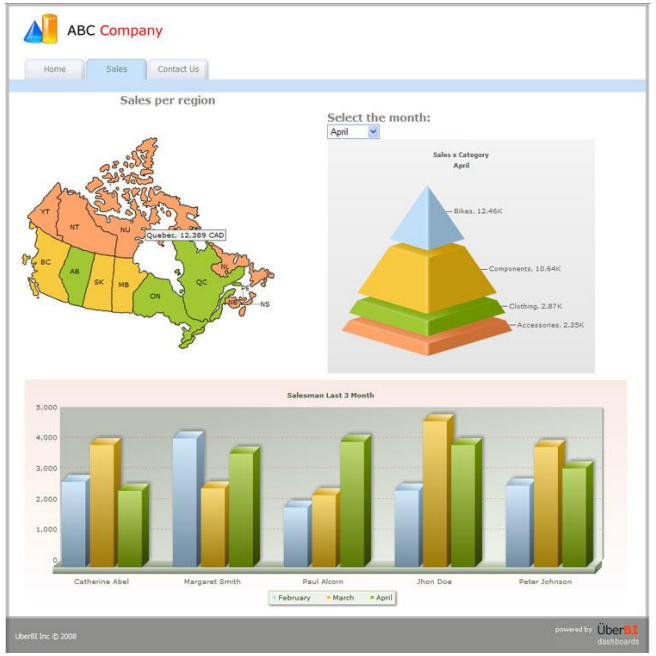
- Reporting: no ICT skills required
- OLAP: users must be able to reason according to the multidimensional paradigm, and they must be acquainted with the visual interface of the tool

Reporting

BUSINESS PERSPECTIVE
Select View
Corporate
Regional
Corporate Revenue
Regional Performance
Category Analysis
Subcategory Analysis
Books
Electronics
Movies
Music
Subcategory
Revenue Forecast
Units Sold

2002 Sales (totals)
$48,191.00
Product 1
Product 2
Product 3
$16,200.00
$15,912.00
Trends
$16,079.00
$15,912.00
$16,200.00
Product 1
Product 2
Product 3
2002 Averages Per Transaction
Average Quantity
Average Amount


Tonnage
50M
350M
World Steel Trade
2004 - 2008 Top Importers and Exporters
InetSoft
Country
Year
Direction
Export
Import


ABC Company
Home
Sales
Contact Us
Sales per region
Select the month:
April
Salesman Last 3 Month
Catherine Abel
Margaret Smith
Paul Alcorn
Jhon Doe
Peter Johnson
February
March
April

OLAP

OLAP is the main way to exploit information in a data warehouse

It gives end-users, whose analysis needs are not easy to define beforehand, the opportunity to analyze and explore data interactively on the basis of the multidimensional model

While users of reporting tools play a passive role, OLAP users are able to actively carry out a complex analysis session, where each step is the result of the outcome of preceding steps

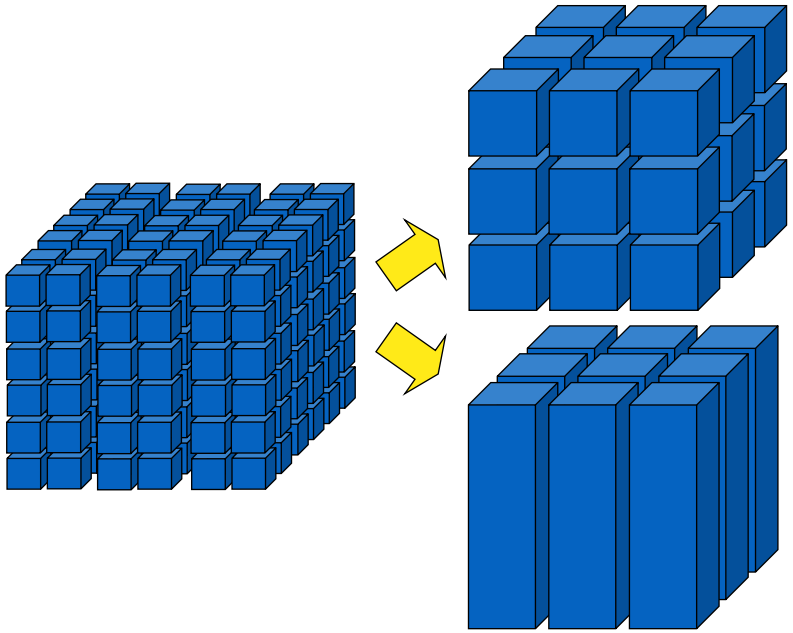
OLAP session

An OLAP session consists of a navigation path that corresponds to an analysis process for facts according to different viewpoints and at different detail levels. This path is turned into a sequence of queries, which are not issued directly, but differentially expressed with reference to the previous query

Every step of an analysis session is characterized by an OLAP operator that turns the latest query into a new one

The results of queries are multidimensional; OLAP tools typically use tables to display data, with multiple headers, colors, and other features to highlight data dimensions

OLAP operators



roll-up

OLAP operators

Metrics / Customer Region / Month	Dollar Sales										
	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Jan 97	$ 620	$ 753	$ 30	$ 660	$ 2.405	$ 1.312	$ 440	$ 1.002	$ 1.002	$ 383	$ 210
Feb 97	$ 258	$ 252	$ 800	$ 975	$ 160	$ 582	$ 744	$ 310	$ 799	$ 118	$ 357
Mar 97	$ 648	$ 244	$ 148	$ 250	$ 1.085	$ 2.961	$ 650	$ 1.240	$ 119	$ 142	$ 96
Apr 97	$ 787	$ 588	$ 447	$ 486	$ 226	$ 506	$ 601	$ 119	$ 550	$ 85	
May 97	$ 1.350	$ 245	$ 936	$ 159	$ 664	$ 626	$ 107	$ 135	$ 200	$ 177	$ 230
Jun 97	$ 842	$ 582	$ 1.281	$ 937	$ 240	$ 774	$ 176	$ 1.139	$ 652	$ 254	$ 745
Jul 97	$ 652	$ 690	$ 486	$ 1.293	$ 605	$ 303	$ 818	$ 103	$ 124	$ 173	$ 66
Aug 97	$ 1.783	$ 304	$ 1.032	$ 170	$ 398	$ 356	$ 432	$ 190	$ 241	$ 407	$ 259
Sep 97	$ 581	$ 778	$ 3.558	$ 587	$ 440	$ 1.652	$ 1.071	$ 315	$ 210	$ 202	
Oct 97	$ 2.291	$ 1.840	$ 600	$ 656	$ 1.300	$ 718	$ 1.210	$ 427	$ 220	$ 520	$ 65
Nov 97	$ 39	$ 1.602	$ 1.082	$ 1.187	$ 842	$ 759	$ 745	$ 232	$ 101	$ 1.037	$ 37
Dec 97	$ 381	$ 1.588	$ 343	$ 118	$ 1.459	$ 635	$ 2.021	$ 259	$ 210	$ 119	$ 189
Jan 98	$ 311	$ 1.174	$ 2.634	$ 3.130	$ 954	$ 2.083	$ 1.351	$ 747	$ 426	$ 447	$ 1.141
Feb 98	$ 2.518	$ 702	$ 1.123	$ 1.336	$ 1.227	$ 3.887	$ 545	$ 268	$ 277	$ 282	
Mar 98	$ 2.459	$ 1.523	$ 1.178	$ 4.708	$ 1.420	$ 3.514	$ 1.948	$ 1.705	$ 276	$ 1.168	$ 63
Apr 98	$ 407	$ 841	$ 524	$ 712	$ 133	$ 2.486	$ 49	$ 390	$ 1.298	$ 221	$ 46
May 98	$ 667	$ 1.721	$ 440	$ 148	$ 80	$ 1.310	$ 303	$ 104	$ 657	$ 65	
Jun 98	$ 699	$ 1.096	$ 898	$ 353	$ 902	$ 839		$ 230	$ 155	$ 105	$ 75
Jul 98	$ 586	$ 1.897	$ 412	$ 226	$ 406	$ 361	$ 1.628	$ 267	$ 1.011	$ 41	$ 184
Aug 98	$ 894	$ 326	$ 792	$ 1.832	$ 1.199	$ 295	$ 1.816	$ 277	$ 102	$ 118	$ 115
Sep 98	$ 338	$ 3.179	$ 505	$ 427	$ 99	$ 2.976	$ 885	$ 135	$ 85	$ 1.110	$ 510
Oct 98	$ 544	$ 413	$ 1.467	$ 209	$ 679	$ 706	$ 556	$ 480	$ 485	$ 99	$ 160
Nov 98	$ 671	$ 459	$ 1.471	$ 2.066	$ 701	$ 716	$ 986	$ 1.127	$ 154	$ 440	$ 361
Dec 98	$ 836	$ 2.096	$ 1.726	$ 3.642	$ 395	$ 1.740	$ 1.943	$ 1.143	$ 366	$ 307	$ 118

roll-up

Metrics / Customer Region / Quarter	Dollar Sales										
	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Q1 1997	$ 1.526	$ 1.249	$ 978	$ 1.885	$ 3.650	$ 4.855	$ 1.834	$ 2.552	$ 1.920	$ 643	$ 663
Q2 1997	$ 2.979	$ 1.415	$ 2.664	$ 1.582	$ 1.130	$ 1.906	$ 884	$ 1.393	$ 1.402	$ 516	$ 975
Q3 1997	$ 3.016	$ 1.772	$ 5.076	$ 2.050	$ 1.443	$ 2.311	$ 2.321	$ 608	$ 575	$ 782	$ 325
Q4 1997	$ 2.711	$ 5.030	$ 2.025	$ 1.961	$ 3.601	$ 2.112	$ 3.976	$ 918	$ 531	$ 1.676	$ 291
Q1 1998	$ 5.288	$ 3.399	$ 4.935	$ 9.174	$ 3.601	$ 9.484	$ 3.844	$ 2.720	$ 979	$ 1.897	$ 1.204
Q2 1998	$ 1.773	$ 3.658	$ 1.862	$ 1.213	$ 1.115	$ 4.635	$ 352	$ 724	$ 2.110	$ 391	$ 121
Q3 1998	$ 1.818	$ 5.402	$ 1.709	$ 2.485	$ 1.704	$ 3.632	$ 4.329	$ 679	$ 1.198	$ 1.269	$ 809
Q4 1998	$ 2.051	$ 2.968	$ 4.664	$ 5.917	$ 1.775	$ 3.162	$ 3.485	$ 2.750	$ 1.005	$ 846	$ 639

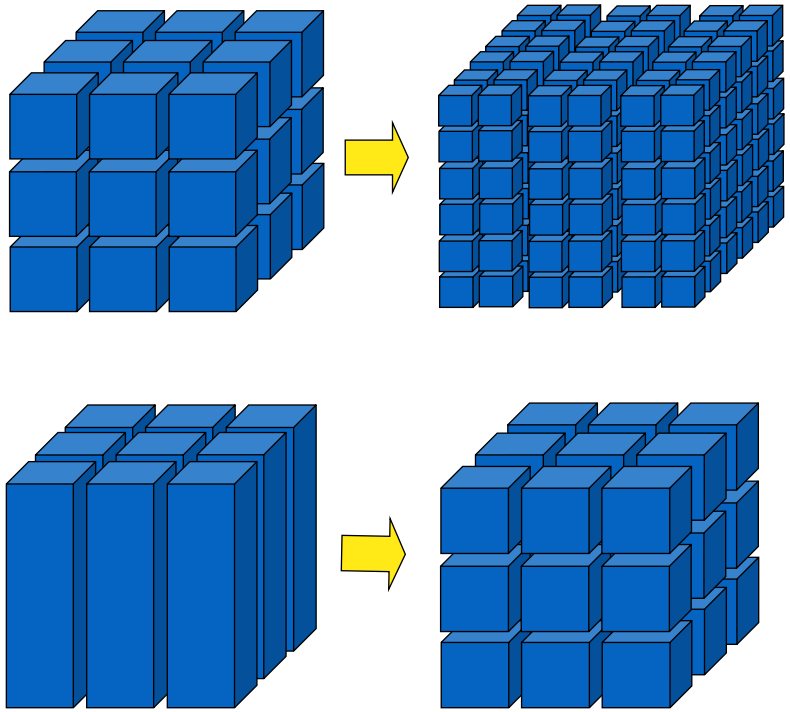
OLAP operators

Category	Year	Metrics / Customer Region	Dollar Sales									
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199	$
	1998		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 7
Food	1997		$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615	$ 1
	1998		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 1
Gifts	1997		$ 2.532	$ 1.355	$ 1.854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375	$ 1.0
	1998		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 6
Health & Beauty	1997		$ 624	$ 640	$ 1.317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292	$ 3
	1998		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72	
Household	1997		$ 5.354	$ 4.112	$ 5.410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875	$ 1.9
	1998		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.7
Kid's Korner	1997		$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 396	$ 105	$ 34	$
	1998		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$
Travel	1997		$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38	
	1998		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$

roll-up

Category	Year / Metrics	Dollar Sales
Electronics	1997	$ 10.616
	1998	$ 29.299
Food	1997	$ 5.300
	1998	$ 5.638
Gifts	1997	$ 16.315
	1998	$ 20.047
Health & Beauty	1997	$ 6.042
	1998	$ 5.665
Household	1997	$ 38.383
	1998	$ 50.391
Kid's Korner	1997	$ 2.559
	1998	$ 2.943
Travel	1997	$ 4.497
	1998	$ 4.792

OLAP operators



drill-down

OLAP operators

Metrics / Customer Region / Quarter	Dollar Sales										
	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Canada
Q1 1997	$ 1.526	$ 1.249	$ 978	$ 1.885	$ 3.650	$ 4.855	$ 1.834	$ 2.552	$ 1.920	$ 643	$ 663
Q2 1997	$ 2.979	$ 1.415	$ 2.664	$ 1.582	$ 1.130	$ 1.906	$ 884	$ 1.393	$ 1.402	$ 516	$ 975
Q3 1997	$ 3.016	$ 1.772	$ 5.076	$ 2.050	$ 1.443	$ 2.311	$ 2.321	$ 608	$ 575	$ 782	$ 325
Q4 1997	$ 2.711	$ 5.030	$ 2.025	$ 1.961	$ 3.601	$ 2.112	$ 3.976	$ 918	$ 531	$ 1.676	$ 291
Q1 1998	$ 5.288	$ 3.399	$ 4.935	$ 9.174	$ 3.601	$ 9.484	$ 3.844	$ 2.720	$ 979	$ 1.897	$ 1.204
Q2 1998	$ 1.773	$ 3.658	$ 1.862	$ 1.213	$ 1.115	$ 4.635	$ 352	$ 724	$ 2.110	$ 391	$ 121
Q3 1998	$ 1.818	$ 5.402	$ 1.709	$ 2.485	$ 1.704	$ 3.632	$ 4.329	$ 679	$ 1.198	$ 1.269	$ 809
Q4 1998	$ 2.051	$ 2.968	$ 4.664	$ 5.917	$ 1.775	$ 3.162	$ 3.485	$ 2.750	$ 1.005	$ 846	$ 639

drill-down

Metrics / Customer City / Quarter	Dollar Sales													
	Arlin	San Pedro	Springfield	Chappel Hill	Scranburg	Pebble Beach	Martinsville	Maddon	Peoria	Pecos	Lake Barkley	Alcameda	Fingers Lake	S
Q1 1997	$ 675										$ 39			
Q2 1997				$ 203					$ 53				$ 135	
Q3 1997				$ 276								$ 252	$ 63	
Q4 1997	$ 215	$ 124			$ 113	$ 45	$ 192	$ 348				$ 79	$ 98	
Q1 1998			$ 140	$ 174			$ 85				$ 237	$ 30	$ 119	
Q2 1998								$ 12	$ 17					
Q3 1998	$ 734					$ 25	$ 1.535							
Q4 1998						$ 219	$ 119	$ 142		$ 85	$ 1.533			

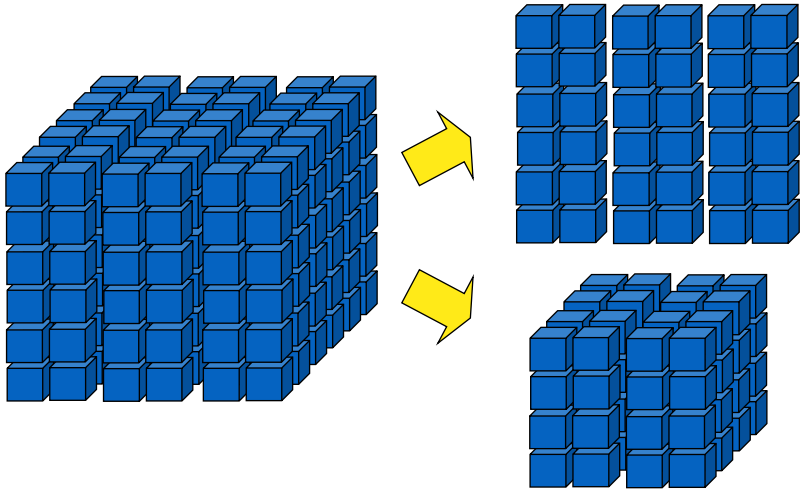
OLAP operators

Metrics	Dollar Sales	
Year	1997	1998
Category		
Electronics	$ 10.616	$ 29.299
Food	$ 5.300	$ 5.638
Gifts	$ 16.315	$ 20.047
Health & Beauty	$ 6.042	$ 5.665
Household	$ 38.383	$ 50.391
Kid's Korner	$ 2.559	$ 2.943
Travel	$ 4.497	$ 4.792

drill-down

Metrics	Dollar Sales											
Customer Region	North-East		Mid-Atlantic		South-East		Central		South		North-West	
Year	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998
Category												
Electronics	$ 138	$ 1.184	$ 1.774	$ 4.529	$ 384	$ 1.892	$ 138	$ 7.232	$ 2.346	$ 651	$ 2.554	$ 9.488
Food	$ 759	$ 538	$ 682	$ 925	$ 729	$ 959	$ 262	$ 677	$ 588	$ 213	$ 469	$ 1.503
Gifts	$ 2.532	$ 1.955	$ 1.355	$ 2.785	$ 1.854	$ 2.800	$ 1.413	$ 2.695	$ 2.535	$ 1.813	$ 2.132	$ 2.844
Health & Beauty	$ 624	$ 611	$ 640	$ 887	$ 1.317	$ 566	$ 647	$ 382	$ 588	$ 499	$ 754	$ 1.162
Household	$ 5.354	$ 5.787	$ 4.112	$ 5.320	$ 5.410	$ 5.416	$ 4.446	$ 6.812	$ 3.058	$ 4.334	$ 3.974	$ 5.008
Kid's Korner	$ 201	$ 247	$ 398	$ 422	$ 485	$ 441	$ 186	$ 380	$ 409	$ 221	$ 323	$ 592
Travel	$ 624	$ 608	$ 505	$ 559	$ 564	$ 1.096	$ 386	$ 611	$ 300	$ 464	$ 978	$ 316

OLAP operators



slice-and-dice

OLAP operators

Category	Year	Metrics / Customer Region	Dollar Sales									
			North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germa
Electronics	1997		$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199	$
	1998		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 7
Food	1997		$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615	$ 1
	1998		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 1
Gifts	1997		$ 2.532	$ 1.355	$ 1.854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375	$ 1.0
	1998		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 6
Health & Beauty	1997		$ 624	$ 640	$ 1.317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292	$ 3
	1998		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72	
Household	1997		$ 5.354	$ 4.112	$ 5.410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875	$ 1.9
	1998		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.7
Kid's Korner	1997		$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 396	$ 105	$ 34	$
	1998		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$
Travel	1997		$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38	
	1998		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$

slice-and-dice

Filter Details:
Year = 1998

Category	Metrics / Customer Region	Dollar Sales										
		North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany	Ca
Electronics		$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462	$ 702	
Food		$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175	$ 100	
Gifts		$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717	$ 686	
Health & Beauty		$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72		
Household		$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649	$ 2.791	$
Kid's Korner		$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19	$ 69	
Travel		$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198	$ 55	

OLAP operators

Metrics	Dollar Sales											
Customer City / Subcategory	Afton	Akron	Albon	Alcameda	Alka	Allagash	Alta	Altoola	Amestra	Amsterdam	Andersonville	Annap
Audio						$ 85						
Automotive								$ 30				
Chocolate	$ 42	$ 42		$ 50		$ 20		$ 22	$ 44			$
Christmas	$ 30					$ 25	$ 30	$ 15				
Classic Toys						$ 7	$ 26				$ 38	
Coffee			$ 9									
Comfort				$ 59		$ 59						
Furniture							$ 485					
Gadgets							$ 199	$ 79	$ 79			
Games & Puzzles							$ 17		$ 45		$ 45	
Gift Baskets			$ 55	$ 43								$
Golf	$ 25							$ 25	$ 14		$ 25	
Hearth									$ 15			
Jewelry	$ 75			$ 189		$ 24	$ 77	$ 189	$ 24			
Kitchen						$ 55	$ 21		$ 76			$
Lawn & Garden	$ 75		$ 100		$ 15	$ 63	$ 100		$ 180	$ 67	$ 40	$
Learning	$ 16							$ 37				
Meat & Cheese		$ 40		$ 20			$ 20				$ 25	
Miscellaneous		$ 200	$ 1.320		$ 200	$ 139			$ 993			
Natural Remedies	$ 13								$ 13			
Pets	$ 215		$ 26			$ 30	$ 68	$ 115	$ 25		$ 34	$
Plants & Flowers	$ 65	$ 65	$ 65				$ 50	$ 60				$
Safety & Security								$ 30	$ 22	$ 22		
Skin Care												
Sleeping			$ 18									
Toys & Accessories							$ 29	$ 185	$ 744			$

slice-and-dice

Filter Details:
Category = Electronics
AND
Dollar Sales > 80
AND
Customer Region = North-West
AND
Year = 1997

Metrics	Dollar Sales					
Customer City / Subcategory	Alta	Armstrong	Avery Heights	Lane	Mt. Everest	San Fransisco
Audio		$ 98		$ 123	$ 85	
Comfort			$ 118		$ 1.495	
Gadgets	$ 199					$ 199

OLAP tools

OLAP tools were born at the end of the 1990s and today continue to renew and include new features

Live time!

Conceptual design

Which formalism?

While it is now universally recognized that a data mart is based on a multidimensional view of data, there is still no agreement on how to implement its conceptual design

Use of the Entity-Relationship model is quite widespread throughout companies as a conceptual tool for standard documentation and design of relational databases, but *it cannot be used to model DWs*

In some cases, designers base their data marts design on the logical level—that is, they directly define star schemata that are the standard ROLAP implementation of the multidimensional model. But a star schema is nothing but a relational schema; *it contains only the definition of a set of relations and integrity constraints!*

The Dimensional Fact Model

The DFM is a graphical conceptual model for data mart design, devised to:

1. lend effective support to conceptual design
2. create an environment in which user queries may be formulated intuitively
3. make communication possible between designers and end users with the goal of formalizing requirement specifications
4. build a stable platform for logical design (*independently of the target logical model*)
5. provide clear and expressive design documentation

The conceptual representation generated by the DFM consists of a set of *fact schemata* that basically model facts, measures, dimensions, and hierarchies

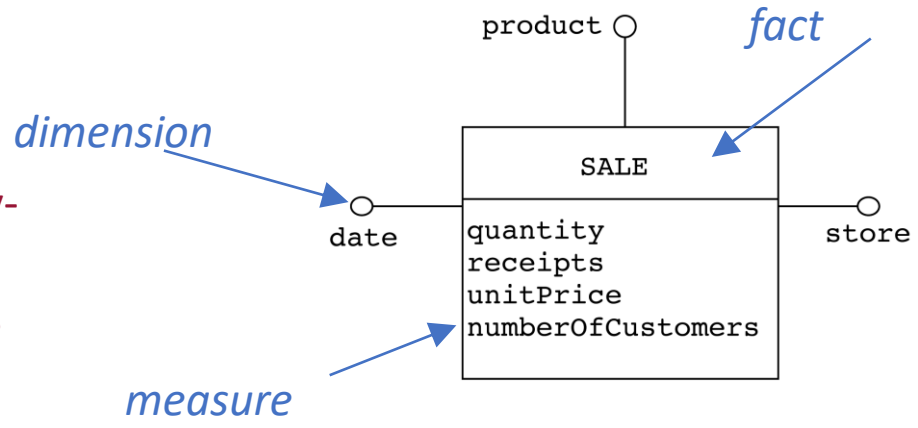
DFM: basic concepts

A fact is a concept relevant to decision-making processes. It typically models a set of events taking place within a company (e.g., sales, shipments, purchases, ...). It is essential that a fact have dynamic properties or evolve in some way over time

A measure is a numerical property of a fact and describes a quantitative fact aspect that is relevant to analysis (e.g., every sale is quantified by its receipts)

A dimension is a fact property with a finite domain and describes an analysis coordinate of the fact. Typical dimensions for the sales fact are products, stores, and dates

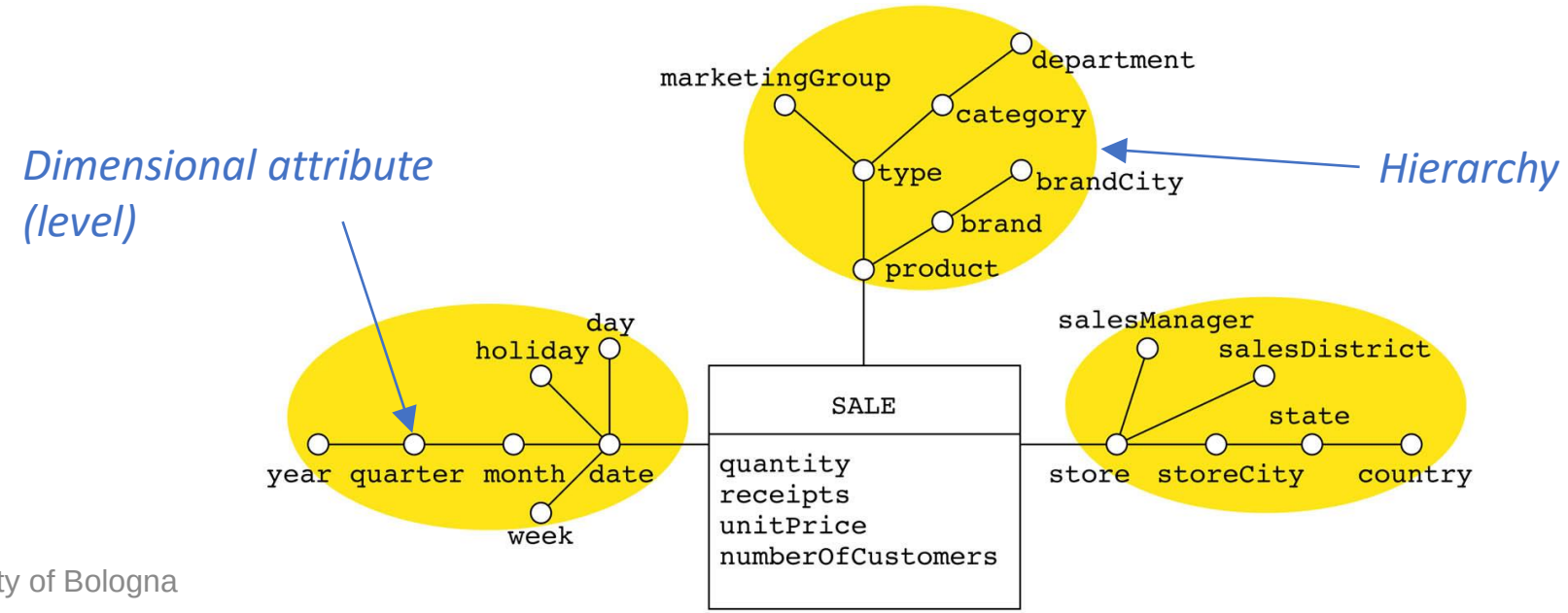
A fact expresses a many-to-many relationship between its dimensions



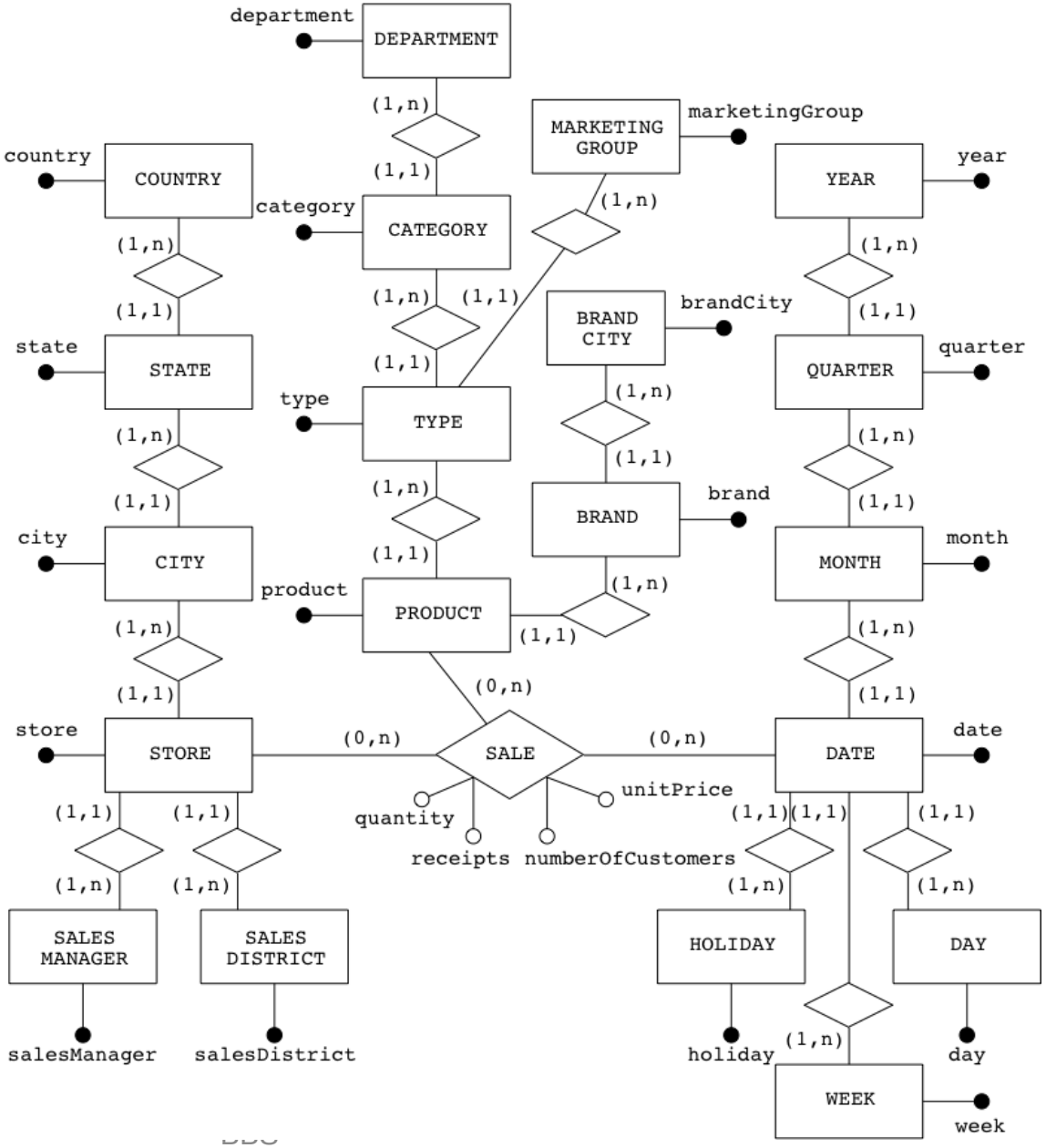
DFM: basic concepts

The general term dimensional attributes stands for the dimensions and other possible attributes, always with discrete values, that describe them (e.g., a product is described by its type, by the category to which it belongs, by its brand, and by the department in which it is sold)

A hierarchy is a directed tree whose nodes are dimensional attributes and whose arcs model many-to-one associations between dimensional attribute pairs. It includes a dimension, positioned at the tree's root, and all of the dimensional attributes that describe it



DFM vs. ERM


department
DEPARTMENT
(1,n)
(1,1)
MARKETING GROUP
marketingGroup
country
COUNTRY
category
CATEGORY
YEAR
year
(1,n)
(1,1)
(1,n)
(1,1)
BRAND CITY
brandCity
state
STATE
type
TYPE
QUARTER
quarter
(1,n)
(1,1)
(1,n)
(1,1)
BRAND
brand
city
CITY
product
PRODUCT
MONTH
month
(1,n)
(1,1)
(1,n)
(1,1)
(0,n)
store
STORE
(0,n)
SALE
(0,n)
DATE
date
unitPrice
quantity
receipts
numberOfCustomers
(1,1)
(1,1)
(1,1)
(1,1)
(1,1)
(1,n)
(1,n)
(1,n)
(1,n)
SALES MANAGER
SALES DISTRICT
HOLIDAY
DAY
salesManager
salesDistrict
holiday
day
(1,n)
WEEK
week